

Specification QLL in MPC

Float structure

Following SecFloat [1, Section V], a floating-point number is parameterised by $p, q \in \mathbb{Z}^+$ ($p = 8, q = 23$ for IEEE single precision [2]) and is a tuple $\alpha = (z, s, e, m)$:

Field	Domain	Meaning
z	$\{0, 1\}$	Zero bit, set iff $\alpha = 0$.
s	$\{0, 1\}$	Sign bit, set only if $\alpha \leq 0$.
e	$\{0, 1\}^{p+2}$, values in $[-2^{p-1} + 1, 2^{p-1}]$	Unbiased signed exponent.
m	$\{0, 1\}^{q+1}$, values in $[2^q, 2^{q+1} - 1] \cup \{0\}$	Normalised unsigned fixed-point mantissa with scale q .

Table 1: Components of a float $\alpha = (z, s, e, m)$.

The tuple represents the real number

$$\alpha = (1 - z) \cdot (1 - 2s) \cdot 2^{\text{int}_{p+2}(e)} \cdot \frac{m}{2^q}$$

where $\text{int}_{p+2}(e)$ is the two's-complement value of e .

A nonzero real x is encoded as

$$e = \lfloor \log_2 |x| \rfloor \quad m = \left\lfloor \frac{|x|}{2^e} \cdot 2^q \right\rfloor$$

(if m overflows to 2^{q+1} , halve it and increment e).

For a float 32 value r $\text{ULP}(r) = 2^{e-23}$ where e is the binary exponent of r i.e. the integer such that $2^e \leq |r| \leq 2^{e+1}$. So the value 2^{e-23} the least significant mantissa bit for a normalized float32 with exponent e .

$$\text{ulp}(r) = 2^{e-23}$$

Examples

$\pi \approx 3.1415927$: $e = \lfloor \log_2 \pi \rfloor = 1$ and $m = \lfloor \frac{\pi}{2} \cdot 2^{23} \rfloor = 13176795$, so

$$\alpha = (0, 0, 1, 13176795)$$

36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0	1	0	0	1	0	0	0	0	1	1	1	1	1	1	0	1	1	0	1	1
z		s		e (10 bits)										m (24 bits)																					

= 0x001C90FDB

$$1 \times +1 \times 2^1 \times 1.5707964 = 3.1415927$$

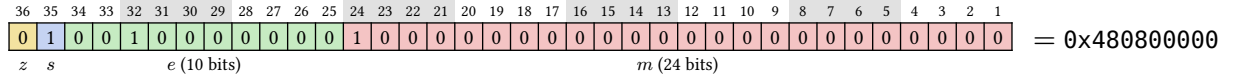
Special values $\pm\infty$

$$(-1)^s \infty = (0, s, 2^{p-1}, 2^q)$$

36	35	34	33	32	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
z		s		e (10 bits)										m (24 bits)																					

= 0x080800000

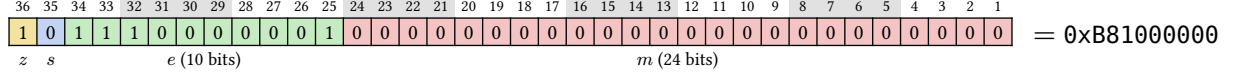
$$1 \times +1 \times 2^{128} \times 1 = +\infty$$



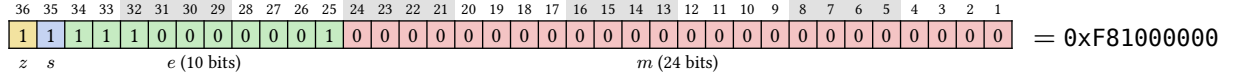
$$1 \times -1 \times 2^{128} \times 1 = -\infty$$

Special values ± 0

$$(-1)^s 0 = (1, s, -2^{p-1} + 1, 0)$$



$$0 \times +1 \times 2^{-127} \times 0 = +0$$



$$0 \times -1 \times 2^{-127} \times 0 = -0$$

NaNs and denormals are not supported.

QLL Operators

Extended Primitive Operations

FUNCTIONALITY $\mathcal{F}_{\text{IntParity}}^{23}(\beta)$

- 1 $e = 1\{\beta.e < 0\} ? 0 : \min(\beta.e, q + 1)$
- 2 $S = L_{\text{pow}2} e$
- 3 $s = \beta.m *_{q+1} S$
- 4 $\text{isInt} = (\neg c \wedge 1\{s \bmod 2^{23} = 0\}) \parallel \beta.z$
- 5 $\text{isOdd} = \text{isInt} \wedge 1\{s \geq 2^{23}\}$
- 6 **Return** isInt, isOdd

Figure 1: Integer parity test for β , used by $\mathcal{F}_{\text{FP}^{\text{pow}}}^{8,23}$.
 The function $L_{\text{pow}2}$ is computed through the LUT operator.

Option 1:

$$\begin{aligned} a &= 2_2^{\log}(a) \\ a^n &= \left(2_2^{\log}(a)\right)^n \\ &= 2^{n \cdot \log_2(a)} \end{aligned}$$

Option 2:

$$\begin{aligned} a &= e^{\ln(a)} \\ a^n &= \left(e^{\ln(a)}\right)^n \\ &= e^{n \cdot \ln(a)} \end{aligned}$$

with $\ln(x) = \log_2 x \cdot \ln 2$

FUNCTIONALITY $\mathcal{F}_{\text{FPpow}}^{8,23}(\alpha, \beta)$

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1  (isInt, isOdd) =  $\mathcal{F}_{\text{IntParity}}(\beta)$ 
2   $s = \alpha.s \ \& \ \text{isOdd}$ 
3  if  $1\{\beta.z = 1\}$  then Return  $\text{Float}_{p,q}(1)$ 
4  else if  $1\{\alpha.z = 1\}$  then
5      if  $1\{\beta.s = 0\}$  then Return  $(1, s, 1 - 2^{p-1}, 0)$  else Return  $(0, s, 2^{p-1}, 2^q)$ 
6  else
7       $\alpha' = (0, 0, \alpha.e, \alpha.m)$ 
8       $\delta = \beta \boxtimes_{p,q} \mathcal{F}_{\text{FPlog2}}^{8,23}(\alpha')$ 
9       $\gamma = \mathcal{F}_{\text{FPexp2}}^{8,23}(\delta)$ 
10 Return  $(\gamma.z, s, \gamma.e, \gamma.m)$ 

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Figure 2: Floating-Point power: α^β

Error propagation:

We want to compute the error propagation of the expression $\hat{y} = \text{FPexp2}(\text{FPmul}(b, \text{FPlog2}(a)))$ and $y = 2^t = a^b$ the expected real result. We recall that $\log_2(a) < 1$ ULP, $b * - < 0.5$ ULP and finally $2^- < 1$ ULP. The ULP for the FPLog2 function can be computed as follows $\text{ulp}(\log_2(x)) = 2^{\lfloor \log_2 \lfloor \log_2(x) \rfloor - 23}$. Let $\ell = \log_2(a)$. The FPLog2 functionality has an error of < 1 ULP, so if $\varepsilon_{\log}$ denotes the error of FPLog2,

$$|\varepsilon_{\log}| < 2^{E_\ell - 23} \quad \text{with} \quad E_\ell = \lfloor \log_2(|\ell|) \rfloor$$

After multiplication by b , this error becomes $b\varepsilon_{\log}$, and the floating-point multiplication adds its own rounding error $\varepsilon_\times$, with

$$\varepsilon = b\varepsilon_{\log} + \varepsilon_\times \quad \text{where} \quad \varepsilon_\times < 0.5$$

Now let $t = b\ell$, since FPexp2 receives $t + \varepsilon$ instead of t , the value before the final floating-point rounding is

$$2^{t+\varepsilon} = 2^t 2^\varepsilon = y 2^\varepsilon.$$

Hence, the absolute difference $y 2^\varepsilon - y = y(2^\varepsilon - 1) \approx \ln(2)|\varepsilon|$. Thus, including the final rounding error ε_{exp} introduced by FPexp2:

$$\begin{aligned}
 |\hat{y} - y| &\leq y |2^{b\varepsilon_{\log} + \varepsilon_\times} - 1| + |\varepsilon_{\text{exp}}| \\
 &< y |2^\varepsilon - 1| + \text{ULP}(2^{t+(b\varepsilon_{\log} + \varepsilon_\times)}) \\
 &\approx y \ln(2)|\varepsilon| + \text{ULP}(2^{t+(b\varepsilon_{\log} + \varepsilon_\times)})
 \end{aligned}$$

Logical disjunction $\vee^\rho$

$$\begin{aligned}
 \llbracket a \vee^\rho b \rrbracket &= a \oplus^p b = (a^p + b^p)^{\frac{1}{p}} \\
 &= M \left(1 + \left(\frac{m}{M} \right)^\rho \right)^{\frac{1}{p}}
 \end{aligned}$$

where

$$M = \max(a, b) \quad m = \min(a, b)$$

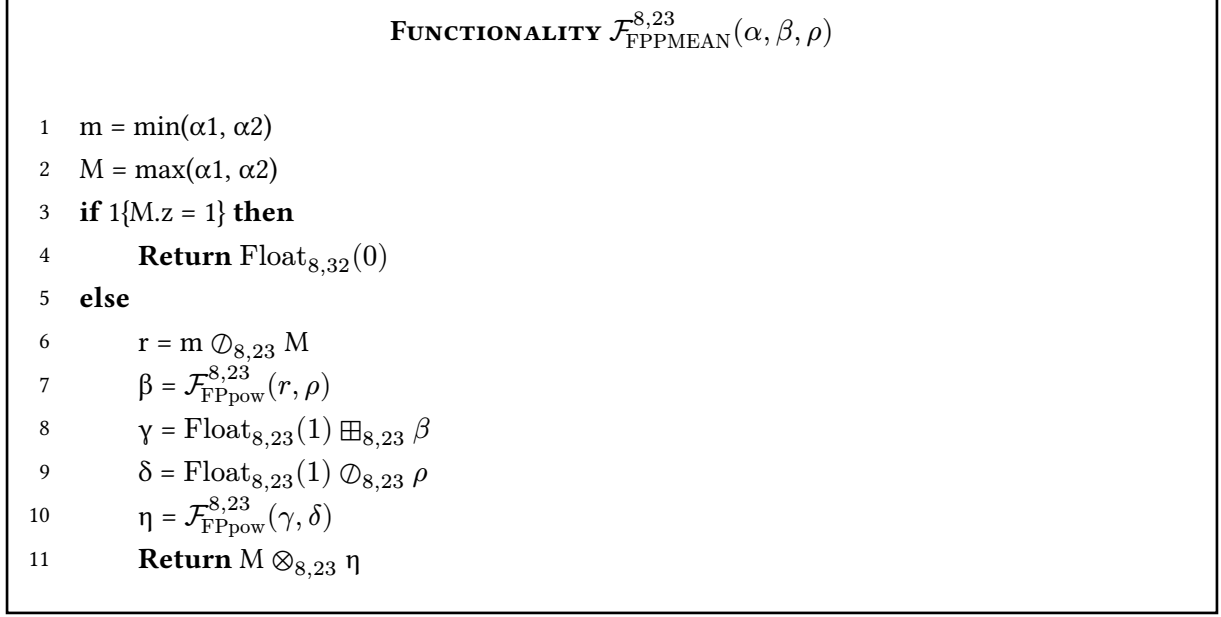


Figure 3: Floating-Point ρ -mean equivalent to $\vee^\rho$

Correctness: TODO

Logical conjunction $\wedge^\rho$

$$\begin{aligned} \llbracket a \wedge^\rho b \rrbracket &= a \oplus^{-\rho} b = (a^{-\rho} + b^{-\rho})^{-\frac{1}{\rho}} \\ &= m \left(1 + \left(\frac{m}{N} \right)^\rho \right)^{-\frac{1}{\rho}} \end{aligned}$$

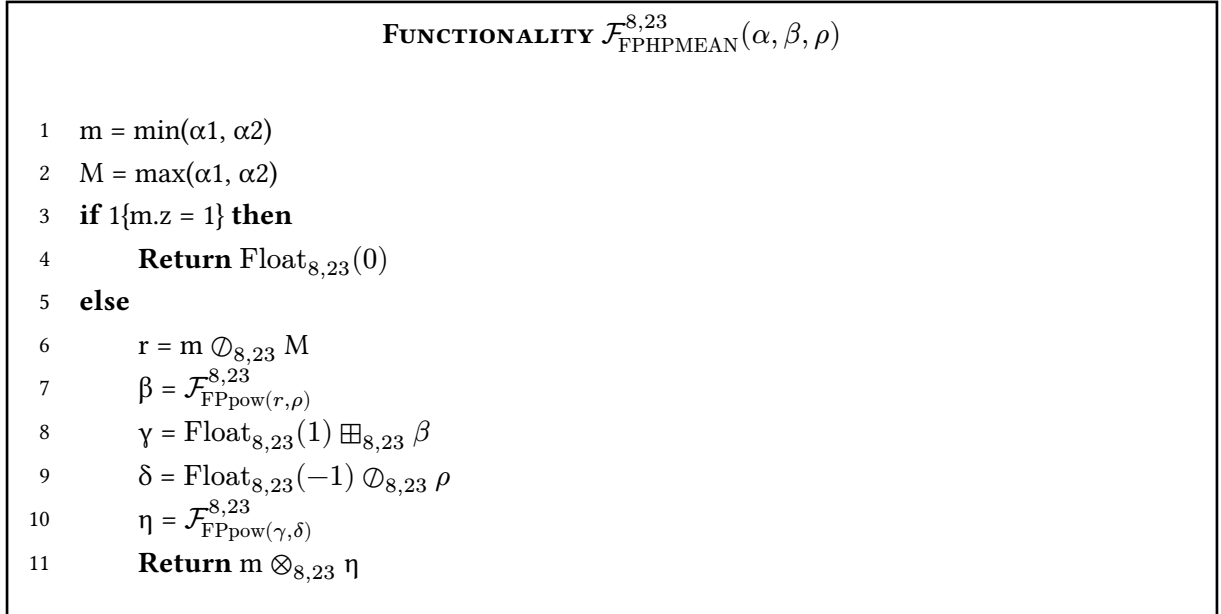


Figure 4: Floating-Point Harmonic ρ -mean equivalent to $\wedge^\rho$

Correctness: TODO

Bibliography

- [1] D. Rathee, A. Bhattacharya, R. Sharma, D. Gupta, N. Chandran, and A. Rastogi, “SecFloat: Accurate Floating-Point meets Secure 2-Party Computation.” [Online]. Available: <https://eprint.iacr.org/2022/322>
- [2] “IEEE Standard for Floating-Point Arithmetic,” *IEEE Std 754-2019 (Revision of IEEE 754-2008)*, vol. , no. , pp. 1–84, 2019, doi: 10.1109/IEEESTD.2019.8766229.